UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 11 Physics Final Exam Practice

Term 1 2026–2027 • Set A • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. What angular displacement equals one complete revolution? [4 marks]

A. π rad
B. π/2 rad
C. 2π rad
D. 4π rad

Reference: pp. 3–4 | **Answer C**

2. For a point at radius r on a rotating disk, tangential speed is: [4 marks]

A. $v = r\omega^2$
B. $v = r\omega$
C. $v = \omega/r$
D. $v = r/\omega$

Reference: p. 5 | **Answer B**

3. A truck has linear acceleration 1.85 m/s² and wheel angular acceleration 5.23 rad/s². What is the wheel radius? [4 marks]

A. 2.83 m
B. 0.177 m
C. 0.354 m
D. 9.68 m

Reference: p. 6, Practice Problem 3 | **Answer C**

4. A toy rotates at one revolution per minute. What is its frequency? [4 marks]

A. 2π Hz
B. 1 Hz
C. 60 Hz
D. 1/60 Hz

Reference: p. 6, Practice Problem 2 | **Answer D**

5. All points on a rigid rotating wheel share the same: [4 marks]

A. Tangential speed regardless of radius
B. Centripetal acceleration regardless of radius
C. Distance travelled regardless of radius
D. Angular displacement over a given interval

Reference: pp. 4–5 | **Answer D**

6. A 0.25 m wrench must supply 35 N m. If the force is perpendicular, its magnitude is: [4 marks]

A. 560 N
B. 140 N
C. 35 N
D. 8.75 N

Reference: p. 9, Practice Problem 11 | **Answer B**

7. Two 0.30 kg point masses are fixed at the ends of a light 0.66 m baton. Its moment of inertia about its midpoint is: [4 marks]

A. 0.131 kg m²
B. 0.0653 kg m²
C. 0.198 kg m²
D. 0.0327 kg m²

Reference: p. 14, Example Problem 3 | **Answer B**

8. A freely suspended irregular object settles with its centre of mass: [2 marks]

A. Vertically below the suspension point
B. At its highest corner
C. Outside the gravitational field
D. Always above the suspension point

Reference: pp. 18–20 | **Answer A**

9. For static equilibrium of a rigid body, which conditions are required? [4 marks]

A. Both force and torque must be positive
B. Only Στ = 0
C. Only ΣF = 0
D. ΣF = 0 and Στ = 0

Reference: pp. 20–21; p. 24, Check Your Progress 41 | **Answer D**

10. The Coriolis effect is observed when motion is described in: [4 marks]

A. A region with no rotation
B. A rotating reference frame
C. A stationary straight line only
D. Every nonrotating inertial frame only

Reference: pp. 23–24 | **Answer B**

11. During a collision between two balls, their interaction forces are: [4 marks]

A. Always zero
B. Larger on the heavier ball
C. Equal in magnitude and opposite in direction
D. Larger on the lighter ball

Reference: pp. 30–31 | **Answer C**

12. On a force–time graph, the signed area represents: [4 marks]

A. Mass
B. Displacement
C. Impulse
D. Velocity

Reference: pp. 30–31 | **Answer C**

13. Two equal-mass freight cars couple. One approaches at 2.2 m/s while the other is at rest. Their final speed is: [4 marks]

A. 2.2 m/s
B. 0 m/s
C. 4.4 m/s
D. 1.1 m/s

Reference: p. 40, Practice Problem 17 | **Answer D**

14. An airbag reduces the average force during a given momentum change mainly by: [4 marks]

A. Increasing the stopping time
B. Increasing the initial momentum
C. Decreasing the stopping time
D. Eliminating momentum conservation

Reference: p. 33, Scientific Thinking; p. 37, Check Your Progress 11 | **Answer A**

15. A spinning skater pulls in the arms with negligible external torque. What happens? [4 marks]

A. I decreases and ω increases
B. Angular momentum becomes zero
C. Both I and ω increase
D. I increases and ω decreases

Reference: pp. 34–35; p. 37, Check Your Progress 15 | **Answer A**

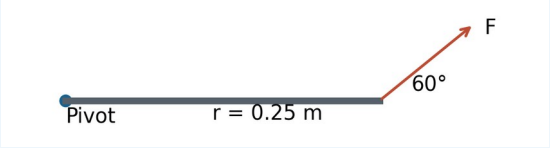
16. A thick heavy rim on a spinning flying disk improves resistance to changes in rotation mainly by increasing its: [2 marks]

A. Moment of inertia
B. Gravitational field strength
C. Linear acceleration in every case
D. Frequency automatically

Reference: p. 47, Check Your Progress 28 | **Answer A**

Set A Section B Free Response

17. A crib toy makes one complete counterclockwise turn per minute. Separately, a bolt requires 35 N m of torque using a 0.25 m wrench pulled at 60.0° to its handle. Take counterclockwise as positive. [10 marks]



Reference: p. 6, Practice Problem 2; p. 9, Example Problem 1; pp. 20–21

(a) Find the toy's angular displacement in 3.0 min and its angular velocity. [3]

Answer Δθ = 3(2π) = +6π rad [1]; ω = Δθ/Δt [1] = +2π rad/min [1].

(b) If the toy slows to a stop, what is the sign of its angular acceleration? Explain. [2]

Answer Negative [1], because positive angular velocity decreases toward zero [1].

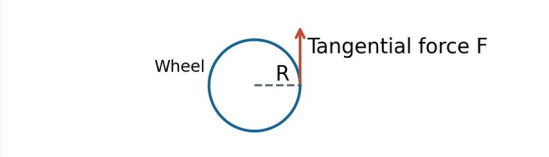
(c) Find the wrench's perpendicular lever arm and the required force. [3]

Answer $r_{perpendicular}$ = 0.25 sin60° = 0.217 m [1]; F = τ/(r sinθ) [1] = 162 N [1].

(d) State both conditions needed for static equilibrium. [2]

Answer ΣF = 0 [1]; Στ = 0 about any axis [1].

18. A solid uniform wheel has mass 15 kg and diameter 0.44 m. It accelerates uniformly from rest to 8.0 rev/s in 15 s. A strap applies a tangential force at its rim. Use $I = \frac{1}{2}MR^2$. [10 marks]



Reference: p. 16, Example Problem 4; pp. 34–36

(a) Calculate its moment of inertia. [2]

Answer R = 0.22 m [1]; $I = \frac{1}{2}(15)(0.22)^2 = 0.363$ kg m² [1].

(b) Find angular acceleration and required torque. [3]

Answer $\omega_f = 16\pi = 50.3$ rad/s [1]; $\alpha = 50.3/15 = 3.35$ rad/s² [1]; $\tau = I\alpha = 1.22$ N m [1].

(c) Find the tangential strap force. [2]

Answer $F = \tau/R$ [1] = 1.22/0.22 = 5.53 N [1].

(d) Calculate the final angular momentum and angular impulse delivered. [3]

Answer $L_f = I\omega_f$ [1] = 18.2 kg m²/s [1]; angular impulse = ΔL = +18.2 N m s [1].

19. A 725 kg car travels east at 115 km/h. The brakes exert a constant westward force of 5.00×10^3 N for 2.00 s. Take east as positive. [10 marks]

Reference: p. 33, Practice Problems 1–2

(a) Convert the initial velocity to m/s and calculate initial momentum. [3]

Answer $v_i = 115/3.6 = +31.94$ m/s [1]; $p_i = mv_i$ [1] = $+2.316 \times 10^4$ kg m/s [1].

(b) Calculate the signed braking impulse. [2]

Answer $J = F\Delta t$ [1] = $(-5000)(2) = -1.00 \times 10^4$ N s [1].

(c) Find final momentum and velocity. [3]

Answer $p_f = p_i + J = +1.316 \times 10^4$ kg m/s [1]; $v_f = p_f/m$ [1] = +18.2 m/s, east [1].

(d) Would a 2175 kg car have to move faster or slower to have the same initial momentum? Find its speed. [2]

Answer Slower, since v = p/m [1]; v = 23159.7/2175 = 10.6 m/s [1].

20. A 1875 kg car travelling at +23 m/s strikes a 1025 kg car travelling at +17 m/s on ice. They stick together. Take the road direction as positive and neglect external impulse. [10 marks]

Reference: p. 40, Example Problem 3; pp. 30–31

(a) Define the system and explain why momentum is conserved. [2]

Answer Both cars [1]; net external impulse is negligible on the icy surface during impact [1].

(b) Calculate total momentum before the collision. [2]

Answer $p_i = 1875(23) + 1025(17)$ [1] = +60550 kg m/s [1].

(c) Find the common final velocity. [3]

Answer $p_i = (m_1 + m_2)v_f$ [1]; $v_f = 60550/2900$ [1] = +20.9 m/s [1].

(d) Find the impulse on each car and compare their directions. [3]

Answer $J_1 = 1875(20.879 - 23) = -3976$ N s [1]; $J_2 = 1025(20.879 - 17) = +3976$ N s [1]; equal and opposite [1].